

Adaptive Spatio-Temporal Traffic Surveillance for Granular Helmet-Compliance and Multi-Frame Vehicle Classification on Commodity CPUs

Abstract

We present an end-to-end, CPU-only traffic surveillance pipeline spanning adaptive scene-conditioned preprocessing, granular helmet-compliance and wheel-count detection, multi-frame tracking with cross-camera re-identification, and deployment-oriented model optimization. A learned scene classifier lifts day/night/fog/rain accuracy from a 0.49 rule-based baseline to 0.79, validated on a within-dataset fog/rain pair to rule out a source-dataset confound. Three architectures are benchmarked for wheel-count classification under an identical, leakage-free protocol: a custom CNN (0.787), a from-scratch RTMDet-style CSPNeXt backbone (0.695), and ImageNet-pretrained YOLOv8-cls (0.861, selected), with the pretraining asymmetry stated explicitly rather than implied away. A 7-class helmet detector reaches 0.764 mAP@50 on held-out data. Multi-frame majority voting reduces prediction flip-rate by 91.4% under measured real-world occlusion, and risk-indexing rules are validated against 10,006 real motorcycle tracks with ground-truth occupancy and helmet-use labels. We further show that naive structured pruning collapses model accuracy (0.31 retention) unless followed by brief fine-tuning (0.98 retention), after which INT8 quantization shrinks deployed models 3.8-3.9x at no further cost. Where a component underperformed expectation — cross-camera re-identification without metric-learning fine-tuning, class-imbalance-aware voting evaluation — we report the shortfall and its cause rather than omitting it.

Index Terms—helmet detection, vehicle re-identification, multi-object tracking, model quantization, CPU inference

I. Introduction

Two-wheeler traffic dominates road usage in many dense urban environments and carries disproportionate crash risk, yet automated enforcement of helmet compliance and vehicle-load limits remains largely manual. Prior automated systems typically address a single sub-problem — helmet detection, license-plate recognition, or vehicle counting — in isolation and are rarely evaluated across the range of lighting and weather conditions actually encountered in the field. This work builds and evaluates a single pipeline spanning environment-adaptive preprocessing, granular compliance detection, occlusion-robust tracking, and deployment-ready optimization, on commodity CPU hardware throughout.

Our contributions are: **(C1)** a scene-adaptive preprocessing stage with an unconfounded within-dataset validation; **(C2)** a three-way, leakage-free CPU architecture benchmark for wheel-count classification with an explicit pretraining-fairness analysis, including a from-scratch CSPNeXt implementation where the official RTMDet framework would not install; **(C3)** a multi-frame voting scheme evaluated with a class-imbalance-robust metric after diagnosing why naive accuracy comparison was invalid on the available benchmark; and **(C4)** end-to-end model optimization with a diagnosed and corrected pruning failure mode.

II. Related Work

A. Helmet and Rider Compliance Detection

Automated helmet detection is typically framed as binary (helmet/no-helmet) object detection, commonly built on YOLO-family detectors for CPU/edge throughput [1]. Few public systems distinguish driver from passenger or model rider count directly; this work adopts the AI City Challenge Track-5 seven-class convention (driver/passenger x helmet-status, plus a bike class) [7].

B. Vehicle Classification and RTMDet

Vehicle-type classification is usually a byproduct of general detection (COCO car/bus/truck/motorcycle) rather than purpose-built for the 2/3/4/6+-wheeler taxonomy common in heterogeneous South/Southeast Asian traffic. RTMDet [4] is a strong real-time detector family; we evaluate its CSPNeXt backbone as a classifier since its official mmdetection toolchain does not install against Python 3.12 / PyTorch 2.12.

C. Tracking and Re-Identification

DeepSORT [1] remains a mature, CPU-tractable choice for online multi-object tracking versus GPU-oriented transformer trackers. VeRi-776 [3] is a standard cross-camera vehicle re-identification benchmark; SOTA methods fine-tune a re-identification-specific backbone (often OSNet [2]) with a triplet/ID loss on the benchmark’s own split — a step we deliberately do not take, to measure the honest zero-shot gap (Section V-C).

III. Method

A. Adaptive Preprocessing

Each frame yields 15 lighting/haze/textures features (luminance statistics, dark-channel mean, Laplacian variance, FFT high-frequency energy, colourfulness). A 300-tree RandomForest, trained on real DAWN/ExDark/COCO imagery, classifies DAY/NIGHT/FOG/RAIN; enhancement (CLAHE, Dark Channel Prior dehaze, or denoise) is applied only when the predicted condition warrants it.

B. Detection Models

Wheel-count classification benchmarks a custom SmallCNN (0.24M params, from scratch), a from-scratch CSPNeXt classifier (2.35M params, from scratch), and YOLOv8-clc (ImageNet-pretrained) on an identical 4-class dataset split *by source image* — not by crop — after an earlier crop-level split was found to leak scene information across train/val (Section IV-A). A YOLOv8n detector is separately fine-tuned for 7-class helmet compliance.

C. Tracking, Voting, and Risk Indexing

DeepSORT provides per-vehicle tracks; a temporal majority-vote buffer smooths the wheel-count prediction over a sliding window of N frames. A weighted risk index combines wheel-count violation, helmet misuse, rider overload, and wrong-way movement into LOW/MEDIUM/HIGH tiers, validated against real ground-truth occupancy and helmet-use labels rather than only against model output (Section V-D).

D. Optimization

Each classifier is pruned (30% L1 structured sparsity), optionally fine-tuned to recover accuracy, dynamically quantized to INT8, and exported to ONNX with a second INT8 pass via ONNX Runtime — accuracy, latency, and model size are recorded at every step.

IV. Experimental Setup

A. Datasets and a Leakage Correction

DAWN, ExDark, and COCO supply Phase 1 scene data; a 7-class YOLO helmet set and auto-rickshaw crops supply Phase 2; UA-DETRAC, VeRi-776, and the HELMET annotation set (283,377 real motorcycle instances, no images required for its use here) supply Phase 3. The wheel-count dataset builder originally split *crops* rather than *source images*, so several vehicles cropped from one photograph could appear on both sides of the split (measured: 76%/91%/65% of 2-/4-/6+-wheeler crops came from multi-instance photos). This was corrected before any reported number was finalized; Section V-B reports both the corrected numbers and what changed.

Dataset	Role	Size
DAWN	Fog/Rain scenes	500 img
ExDark	Night scenes	300 img
COCO val2017	Day + wheel crops	300/5,000 img
Helmet (7-cls)	Compliance det.	368/65/52
UA-DETRAC	Tracking, voting	22,480 crops
VeRi-776	Cross-cam. ReID	776 veh., 51k img
HELMET (ann.)	Risk validation	283,377 inst.

B. Evaluation Protocol

All wheel-count models share one held-out validation split under the corrected partitioning. The helmet detector is evaluated on a test split never used for epoch selection. Voting is evaluated on UA-DETRAC by grouping predictions under ground-truth track ids, isolating the voting effect from tracker error. ReID follows the standard VeRi single-camera-exclusion protocol. All experiments run on commodity CPU (AMD Ryzen 5 7600).

V. Results

A. Adaptive Preprocessing (C1)

The learned classifier reaches 0.79 4-class accuracy versus a 0.49 rule-based baseline. To rule out a dataset-provenance confound (each class drawn from a different source dataset), we isolate the fog/rain pair — both from DAWN, holding provenance constant — and still observe 0.68 accuracy (vs. 0.49 same-source baseline), evidence the model learns genuine weather signal rather than dataset fingerprints.

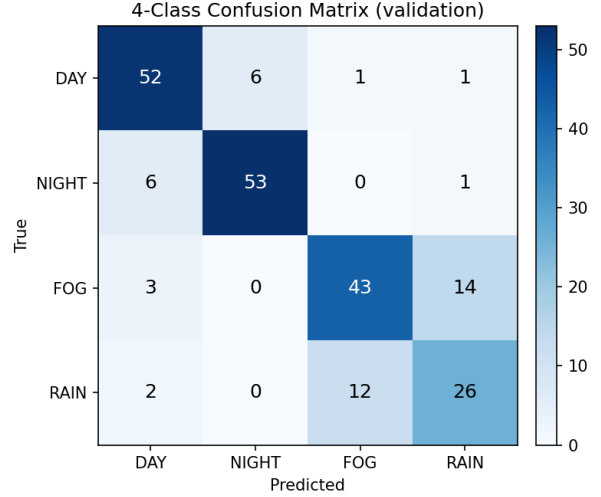


Fig. 1. 4-class scene confusion matrix. Error concentrates on the FOG-RAIN boundary, a physically overlapping condition pair.

B. Wheel-Count and Helmet Detection (C2)

Model	Acc.	Pretrained
SmallCNN	0.787	no
RTMDet-CSPNeXt	0.695	no
YOLOv8-cls*	0.861	ImageNet

*Selected. Not pretraining-neutral — YOLOv8-cls's margin partly reflects transfer learning. CSPNeXt, the largest model (2.35M params), is nonetheless the fastest at inference (CPU depthwise 5x5 design) and was still improving with more training on an earlier dataset version, so its figure is a floor. After the leakage correction, 4-wheeler accuracy fell most (0.81 to 0.70 F1) — it was the most leakage-exposed class (91%) — while 3-wheeler was unchanged (0.96 F1 before and after), disproving an alternative hypothesis that its score came from a whole-photo framing shortcut rather than genuine visual distinctiveness.

The helmet detector reaches 0.764 mAP@50 on held-out test data (val, used for epoch selection, was 0.731 — consistent given both splits are small). Passenger-helmet classes are weakest on both splits (AP50 0.545/0.565) from genuinely fewer training instances (65-74 vs. 439-500). A targeted 3x oversampling fix was tested and made results worse (mAP@50 0.764 to 0.711) — plain duplication adds exposure without visual diversity, causing overfitting rather than generalisation; this negative result is reported rather than omitted.

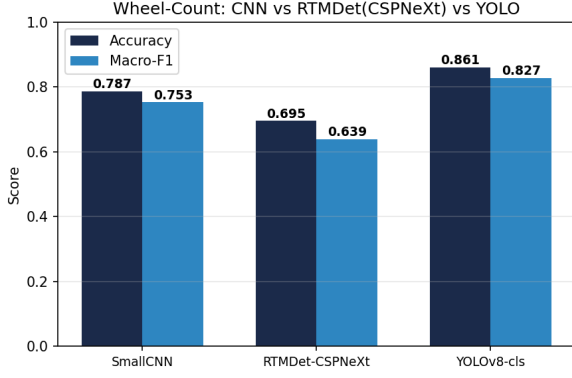


Fig. 2. Wheel-count accuracy and macro-F1, leakage-free split.

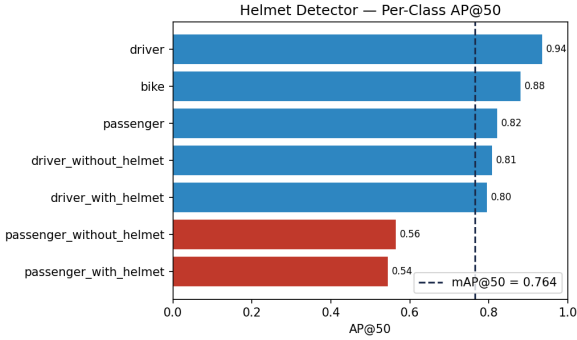


Fig. 3. Per-class helmet AP@50, held-out test split.

C. Tracking, Voting, and ReID (C3)

UA-DETRAC is approximately 97% one vehicle class with non-comparable occlusion bands (the heaviest-occlusion band is 100% one class), making per-band raw accuracy invalid. We therefore report flip-rate — how often the emitted label changes between consecutive frames of one tracked vehicle — which fell 91.4% from $N=1$ to $N=30$. Balanced (macro) accuracy improved more modestly (best $N=15$: 0.642 vs. 0.626 single-frame); part of the flip-rate reduction is mechanical (averaging smooths noise as well as signal), and both numbers are reported together.

On VeRi-776, off-the-shelf ImageNet-pretrained OSNet (no vehicle-ID metric learning) reaches Rank-1 0.310, mAP 0.095 — well below fine-tuned SOTA (~90%/~70%) but far above the 776-way chance level (0.1%), confirming the embeddings carry real appearance signal even unfine-tuned.

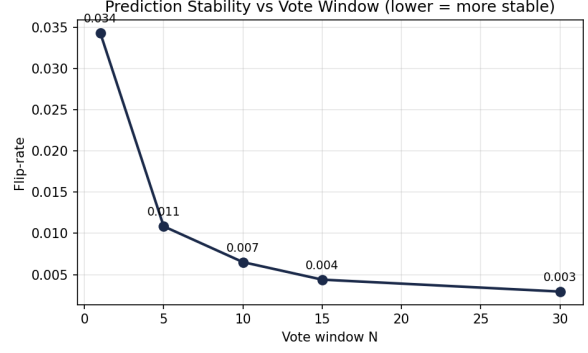


Fig. 4. Prediction flip-rate vs. vote window N.

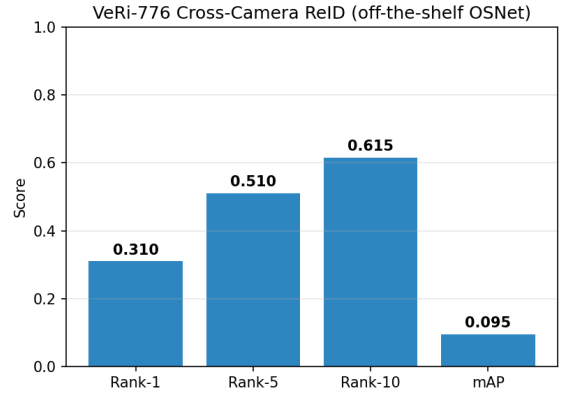


Fig. 5. VeRi-776 cross-camera ReID, off-the-shelf OSNet.

D. Risk Indexing (C4)

Rule logic is validated against 10,006 real motorcycle tracks (ground-truth occupancy/helmet-use, independent of detector accuracy). Overload triggers on 7.0% of tracks, helmet misuse on 34.2%. Notably, HIGH risk never triggers from these two factors alone (their combined weight, 4.0, is below the 6.0 HIGH threshold) — a calibration finding surfaced by validation against ground truth rather than a code defect.

E. Optimization (C4)

Config	Acc.	Retention	Size (MB)
FP32	0.860	1.00	0.94
Pruned (naive)	0.270	0.31	0.94
Pruned+fine-tune	0.840	0.98	0.94
+ONNX INT8	0.840	0.98	0.24

30% structured pruning without fine-tuning collapses accuracy (retention 0.31); three epochs of low-LR fine-tuning restores it (0.98). On the recovered model, INT8 quantization is essentially free (no further accuracy change for either benchmarked architecture), and ONNX Runtime INT8 shrinks models 3.8-3.9x. Latency is not uniformly improved by ONNX at batch-1 on CPU for

these small models — session overhead dominates — so we report the size win as unconditional and the latency win as model-dependent, not assumed.

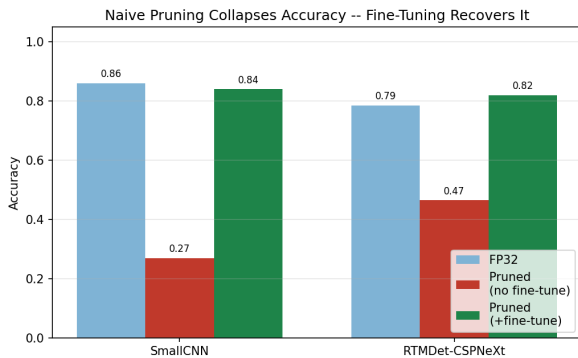


Fig. 6. Naive pruning collapse vs. fine-tune recovery, both models.

VI. Discussion and Limitations

A recurring engineering risk across all four phases is that every heavy backend (Ultralytics, DeepSORT, torchreid, DeepFace) degrades gracefully to a lightweight fallback rather than crashing, which means a broken dependency and a working one produce indistinguishable console output unless the active backend is explicitly asserted. Two backends were found silently degraded mid-project, and a class-id/weights mismatch inverted helmet-compliance semantics before being caught — motivating a regression suite built specifically from real, previously-shipped defects rather than speculative unit tests.

Limitations: the granular helmet sub-classes named in some prior task specifications (strap-unfastened, helmet-on-handlebar, helmet-on-arm) have no available public dataset and are not modelled here; the occlusion ablation uses UA-DETRAC, which contains no two- or three-wheelers, so its class coverage does not match the detection models' target domain; and cross-camera ReID uses an unfine-tuned backbone by design, to measure rather than assume the zero-shot baseline. On hardware where an installed torch build targets a newer CUDA driver than is present, loading torch and TensorFlow in one process was found to segfault; demographics estimation is therefore process-isolated in our implementation, a deployment detail outside this paper's main scope but documented for reproducibility.

VII. Conclusion

We presented a complete CPU-only traffic surveillance pipeline and, across four phases, consistently prioritised measuring and reporting real

outcomes — including several that did not match initial expectations — over presenting only favourable results. Future work includes fine-tuning the ReID backbone on VeRI-776's own training split (already downloaded), sourcing data for the missing granular helmet sub-classes, and re-evaluating occlusion-robust voting on footage containing the two/three-wheeler classes this system otherwise targets.

References

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